# Frozen-Corpus Exhaustion without Architecture Closure: Integrating the Three H1 Structural Roots

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**Abstract**

TPC-173–175 freeze every TPC-133–172 manuscript and the declared data substrate, define a source-locked production local-edge witness contract, and extract the maximal qualifying family. The family is empty in that corpus. TPC-176 sends only proved edges to the TPC-165 gluing interface and obtains an exact empty-domain coverage ledger without asserting totality. TPC-177 shows that active support is vacuous on the empty eligible domain and cannot close its existential root. TPC-178 shows that no physical representation class is eligible for a canonicality or minimality audit and preserves the archive-key firewall.

We integrate these results without turning a scoped source-extraction stop into global architecture infeasibility. The three independent H1 structural roots all remain NOT-TESTABLE; the selected first missing object is the source-backed local occurrence-edge family. The current architecture verdict remains NOT_TESTABLE. The result is an L1 structural integration supported by L0 executable audits, not fixed-phase or fixed-$h_0$ arithmetic progress and not program-positive L2.

## 1 Frozen source result

Let

$$\mathcal{S}_{133:172} = \{\text{the forty source-locked main manuscripts TPC-133–172}\} \tag{1}$$

together with the separately declared data/status substrate. TPC-173 gives every manuscript exactly one disposition and finds no `NOT_MAPPED_YET` file and no qualifying production local actual-occurrence-edge claim [2]. TPC-174 fixes the minimal witness contract, and TPC-175 extracts

$$\mathcal{E}_{175} = \varnothing. \tag{2}$$

The conclusion (2) is maximal in the frozen declared corpus. It is not a theorem that such edges do not exist in mathematics.

The exact scoped cell is

$$\begin{aligned} &\texttt{production\_local\_edge\_extraction\_from\_tpc133\_172},\\ &\text{status} = \mathsf{STOP\_SCOPED},\\ &\text{scope} = \mathcal{S}_{133:172}. \end{aligned} \tag{3}$$

Changing the source corpus changes the instance of (3).

## 2 Three structural roots

The refined H1 sub-DAG has the independent roots

$$\begin{aligned} E &= \text{H1.source\_backed\_local\_occurrence\_edge\_family},\\ A &= \text{H1.actual\_active\_support\_certificate},\\ M &= \text{H1.canonical\_minimal\_representation\_certificate}. \end{aligned} \tag{4}$$

TPC-166 proved that none is an ancestor of another [1]. TPC-173–178 refine their current evidence but do not change this independence.

### 2.1 The local-edge root

The frozen-corpus extraction method for $E$ is stopped by (2). Nevertheless $E$ remains NOT-TESTABLE as an architecture root: a new source-backed theorem or an explicitly enlarged corpus could supply it. A scoped empty inventory is not global impossibility.

### 2.2 The active-support root

TPC-176 produces no eligible carrier. TPC-177 consequently has the exact coefficient ledger

$$(\text{eligible}, \text{nonzero}, \text{zero}, \text{missing}) = (0, 0, 0, 0). \tag{5}$$

The universal active-support sentence is vacuous and supplies no existential production witness. Hence $A$ remains NOT-TESTABLE. This root is also disjoint from the H9 literal-weight registry.

### 2.3 The representation root

TPC-178 forms no eligible physical representation class. The five-field TPC-164 archive key remains an archive address, not a physical selector. Zero tested classes and zero counterexamples prove neither canonicality nor noncanonicality. Thus $M$ remains NOT-TESTABLE.

## 3 Recomputed antichain

**Theorem 3.1** (TPC-179 H1 structural verdict)**.** *For the source-locked TPC-173–178 state, the minimal H1 structural root antichain is*

$$\mathcal{B}_{H1} = \{E, A, M\}. \tag{6}$$

*The current verdict and selected reporting representative are*

$$\begin{aligned} \mathit{current\_verdict} &= \mathsf{NOT\_TESTABLE}, \\ \mathit{first\_missing} &= E. \end{aligned} \tag{7}$$

*Proof.* TPC-175 does not prove $E$; it stops only the scoped extraction cell (3). TPC-177 proves neither an existential active carrier nor $A$. TPC-178 proves neither a physical representation selector nor $M$. The three nodes are parent-free and independent in the refined H1 sub-DAG, so each is minimal and none suppresses another. The declared obligation order selects $E$ without erasing $A$ or $M$. Since at least one selected-root prerequisite is missing and no global architecture infeasibility theorem is present, the only valid verdict is (7). □

| Object | Status | Exact reason |
|---|---|---|
| frozen-corpus extraction cell | STOP-SCOPED | complete declared corpus has zero qualifying claims |
| local-edge root $E$ | NOT-TESTABLE | scoped exhaustion is not mathematical nonexistence |
| active-support root $A$ | NOT-TESTABLE | empty eligible domain has no existential witness |
| representation root $M$ | NOT-TESTABLE | no eligible physical representation class |
| occurrence-augmented architecture | NOT-TESTABLE | three required roots remain missing; no global stop theorem |

## 4 Scoped stops and legal continuations

A route cell is a method with a declared input and scope. An architecture is a complete candidate path to the physical endpoint. TPC-179 records two separate cells:

| cell | type | status | |
|---|---|---|---|
| TPC-133–172 source extraction | SOURCE_EXTRACTION_METHOD | STOP_SCOPED | (8) |
| occurrence-augmented H1 path | ARCHITECTURE_ROUTE | NOT_TESTABLE. | |

The following are valid next actions:

1. enlarge the source corpus explicitly and rerun the inventory, schema, family, coverage, support, and representation chain;
2. prove a new production local-edge theorem satisfying the TPC-174 contract;
3. if a complete architecture is separately proved infeasible, reroute through a fresh source-backed crosswalk rather than renaming the stopped cell.

Silently treating a synthetic witness as production evidence, or calling the architecture stopped because one extraction cell is empty, is invalid.

## 5 Literal fixed-$h_0$ boundary

The physical target retains the literal requirement

$$h_0 = 2. \tag{9}$$

Some source records in the frozen corpus preserve this lineage, but the current structural audit supplies no eligible occurrence carrier to which (9) can be attached. TPC-179 therefore records:

$$\begin{gathered} \text{required physical value} = 2, \\ \text{eligible carrier with fixed-}h_0\text{ lineage} = \text{none}, \\ \text{fixed-}h_0\text{ arithmetic progress claimed} = \text{false}. \end{gathered} \tag{10}$$

The declaration of the required value is a boundary condition, not an arithmetic theorem.

Likewise, TPC-179 consumes no named fixed phase, literal target normalization, deterministic endpoint, or strict endpoint slack. It cannot pay any part of the 1/400 budget.

## 6 Machine interface and regression audit

The export `tpc179_h1_integration.json` is designed for dynamic import by TPC-182. It contains the verdict, first missing node, full antichain, root ledger, scoped route cells, fixed-$h_0$ boundary, source locks, and claim boundary. A separate audit locks its canonical payload.

Mutation regressions reject:

- promotion of the scoped extraction stop to architecture infeasibility;
- removal or reordering of independent roots;
- promotion of empty-domain active support;
- promotion of the archive key to physical canonicality;
- use of $h_0 = 2$ as an arithmetic gain;
- promotion to program-positive L2, strict 1/400, a prime-pair lower bound, or the twin-prime theorem.

All hashes have integrity semantics only.

# 7 Conclusion

The TPC-133–172 production source corpus is exhausted under an explicit witness contract, and its local-edge extraction cell is properly stopped. The architecture is not closed: its three independent structural roots remain NOT-TESTABLE. This sharpened status is the correct input to the later production-phase and metric-to-fixed atom audits. No fixed-phase or fixed-$h_0$ arithmetic conclusion has been promoted.

# References

[1] Liang Wang. A refined h1 crosswalk frontier: Three independent production roots. TPC-166 manuscript, 2026.

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[3] Liang Wang. The maximal source-backed local occurrence family in the frozen production corpus. TPC-175 manuscript, 2026.

[4] Liang Wang. Coverage before gluing: An empty-domain audit for the source-backed local occurrence family. TPC-176 manuscript, 2026.

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[6] Liang Wang. Eligibility before canonicality: Why an archive key is not a physical representative. TPC-178 manuscript, 2026.